Foundations of Machine Learning (IT582)

2nd In-Semester Exam

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Date: 07/10/2026 Total Marks: 50 Duration: 2 Hrs.

All The Best!!

1. There are many criteria for active learning. In particular, in the context of linear regression, we derived such criteria by assuming that the underlying model was also linear (in the feature space). One of the resulting criteria was based on finding points where our predictions varied the most (relative to resampled training sets from an assumed underlying model).

We will focus here on simple active learning methods for classification tasks with the perceptron algorithm. We assume that you can only ask labels for the training examples $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ (those we don't already have labels for). The labels are fixed once revealed so there's no reason to query the same point multiple times. The perceptron algorithm, in response to mistakes, updates its parameters according to

$$\theta \leftarrow \theta + y_t \mathbf{x}_t \quad \text{iff} \quad y_t \theta^T \mathbf{x}_t \leq 0$$

(a) In our setting, would it be useful to get a label for a point N that we can classify correctly? Justify your answer.

(b) Given the current θ we have to select which example $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ would be the most useful to label. Select all of the following criteria you believe would work as a selection criterion. We could select the point $\mathbf{x}_t$ with

- the largest norm $||\mathbf{x}_t||$
- the largest value of $|\theta^T \mathbf{x}_t|$
- the smallest value of $|\theta^T \mathbf{x}_t|$

Briefly explain why your chosen criterion (criteria) would work in our active learning setting.

[Marks 8]

2. Consider the following two-class dataset, where each sample is represented by two features (x_1, x_2):

Sample	x_1	x_2	Class
A	1	0	P
B	0	2	P
C	2	0	N
D	0	3	N
E	4	0	N
F	0	5	P
G	6	0	N

A new sample is given by

$$x = (0, 0).$$

Use Euclidean distance for all calculations.

(a) Calculate the Euclidean distance of the new sample from each of the training samples and arrange the samples in increasing order of distance.

(b) Using $K = 3$, classify the new sample using the majority voting rule.

(c) Repeat the classification using $K = 5$. Does the predicted class change? Briefly explain why changing K can affect the classification result.

(d) Suppose that instead of giving equal votes to the K neighbours, each neighbour is assigned a weight

$$w_i = \frac{1}{d_i},$$

where d_i is its distance from the test sample. Using the five nearest neighbours, determine the predicted class under this distance-weighted K-NN rule.

[Marks 8]

3. Consider a binary classification problem for classifying emails as either *Spam* or *Not Spam*. Each email is represented using three binary features:

$$X_1 = \text{contains "free"}, \qquad X_2 = \text{contains "offer"}, \qquad X_3 = \text{contains "meeting"}.$$

The following training data are available:

Email	X_1	X_2	X_3	Class
1	1	1	0	Spam
2	1	0	0	Spam
3	1	1	0	Spam
4	0	1	0	Spam
5	0	0	1	Not Spam
6	0	1	1	Not Spam
7	0	0	1	Not Spam
8	1	0	1	Not Spam

Using the Naive Bayes classifier, classify the new email

$$x = (X_1, X_2, X_3) = (1, 0, 1).$$

(a) Classify the new email using the standard Naive Bayes classifier **without smoothing**. Show the calculation of the class scores and identify the predicted class.

(b) Repeat the classification using **Laplace smoothing**. Show the updated class scores and identify the predicted class. Briefly explain why the result differs from part (a), if applicable.

[Marks 10]

4. Let $x^{(1)}, x^{(2)}, \ldots, x^{(n)}$ be n independent observations generated according to

$$x^{(i)} = A + w^{(i)}, \qquad i = 1, \ldots, n,$$

where A is an unknown deterministic quantity and $w^{(i)} \sim \mathcal{N}(0, \sigma^2)$ are independent and identically distributed random variables.

(a) Find the maximum-likelihood estimator (MLE) of the parameter A using all n observations. Show that MLE $\hat{A}$ is unbiased and derive its variance.

(b) In Leave-One-Out Cross-Validation (LOOCV), the i-th observation is excluded while estimating A. Thus, the estimator $\hat{A}_{-i}$ obtained after leaving out $x^{(i)}$. Show that $\hat{A}_{-i} = A$. What does happen to the variance of the LOOCV estimator.

(c) Define the LOOCV prediction error for the i-th observation as

$$e^{(i)} = x^{(i)} - \hat{A}_{-i}.$$

Using

$$x^{(i)} = A + w^{(i)}$$

and the expression for $\hat{A}_{-i}$, derive $e^{(i)}$ explicitly in terms of $w^{(i)}$ and the remaining noise variables.

(d) Show that the LOOCV prediction error is unbiased and show its variance

$$\boxed{\text{Var}(e^{(i)}) = \frac{n}{n-1}\sigma^2}.$$

[Marks 12]

5. We consider here a discriminative approach for solving the classification problem illustrated in Figure 1.

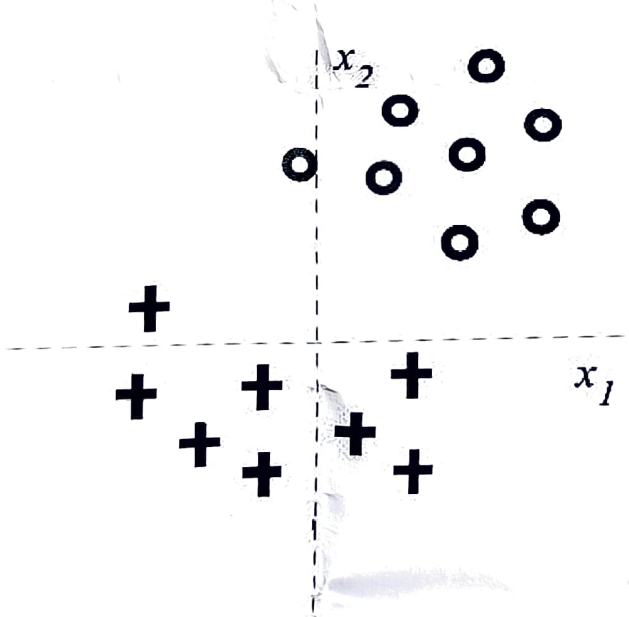


Figure 1: The 2D labeled training set, where '+' corresponds to class $y = 1$ and 'O' corresponds to class $y = 0$.

Part 1: We attempt to solve the binary classification task depicted in Figure 1 with simple logistic regression model.

$$P(y = 1|\mathbf{x}, \boldsymbol{\theta}) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2) = \frac{1}{1 + \exp(-\theta_0 - \theta_1 x_1 - \theta_2 x_2)}$$

Notice that the training data can be separated with zero training error with a linear separator. Consider training regularized linear logistic regression models where we try to maximize

$$\sum_{i=1}^{n} \log(P(y_i|x_i, \theta_0, \theta_1, \theta_2))) - C\theta_j^2$$

for large C. The regularization penalties used in penalized conditional log-likelihood estimation are $-C\theta_j^2$, where $j = \{0, 1, 2\}$. In other words, only one of the parameters is regularized in each case. Given the training data in Figure 1, how does the training error change with regularization of each parameter θ_j? State whether the training error increases or stays the same (zero) for each θ_j for very large C. Provide a brief justification for your answers.

Part 2: If we change the form of regularization to $L1$ norm and regularize θ_1 and θ_2 only(but not w_0), we get the following penalized log-likelihood.

$$\sum_{i=1}^{n} \log(P(y_i|x_i, \theta_0, \theta_1, \theta_2))) - C(|\theta_1| + |\theta_2|)$$

As we increase the regularization parameter C which of the following scenarios do you expect to observe? (Choose only one) Briefly explain your choice.

- *First θ_1 will become 0, then θ_2*
- *First θ_2 will become 0, then θ_1*
- *θ_1 and θ_2 will become zero simultaneously.*
- *None of the weights will become exactly zero, only smaller as C increases.*

(a) For very large C, with the same $L1$-norm regularization for θ_1 and θ_2 as above, which value(s) do you expect θ_0 to take? Explain briefly. (Note that the number of points from each class is the same.) (You can give a range of values for θ_0 if you deem necessary).

(b) Assume that we obtain more data points from the '+' class that corre- sponds to $y = 1$ so that the class labels become unbalanced. Again for very large C, with the same $L1$-norm regularization for θ_1 and θ_2 as above, which value(s) do you expect θ_0 to take? Explain briefly. (You can give a range of values for θ_0 if you deem necessary).

[Marks 12]